

C++ Inheritance – Practice Task Sheet

First 20 Programming Tasks | Beginner to Real-World Projects

Task 1: Single Inheritance – Student & Person

Create a Person class with name and age. Create a Student class derived from Person with roll number and marks. Display all student details using a Student object.

Task 2: Single Inheritance – Employee & Manager

Create an Employee class with name and salary. Derive a Manager class with department and bonus. Calculate and display the manager's total salary.

Task 3: Single Inheritance – Vehicle & Car

Create a Vehicle class with brand and speed. Derive a Car class with model and price. Display complete car details.

Task 4: Single Inheritance – Animal & Dog

Create an Animal class with name and age. Derive a Dog class with breed. Display complete dog details.

Task 5: Multilevel Inheritance – Person → Student → Result

Create Person, Student and Result classes using multilevel inheritance. Store name, age, roll number, course, marks and percentage, then display the complete result.

Task 6: Multilevel Inheritance – Vehicle → Car → SportsCar

Create Vehicle, Car and SportsCar classes using multilevel inheritance. Store brand, speed, model, price and horsepower, then display all details.

Task 7: Multilevel Inheritance – Employee → Developer → SeniorDeveloper

Create Employee, Developer and SeniorDeveloper classes. Store name, salary, programming language and experience. Display complete senior developer details.

Task 8: Multiple Inheritance – Student + Sports → Result

Create Student and Sports classes and derive a Result class from both. Store academic marks and sports score, then calculate and display the total score.

Task 9: Multiple Inheritance – Academic + Sports → Student

Create Academic and Sports classes and derive a Student class from both. Combine academic and sports marks to calculate and display the final result.

Task 10: Multiple Inheritance – Teacher + Researcher → Professor

Create Teacher and Researcher classes and derive a Professor class from both. Store subject, research area and experience, then display professor information.

Task 11: Hierarchical Inheritance – Animal → Dog, Cat, Cow

Create an Animal base class and derive Dog, Cat and Cow classes from it. Add unique data or behavior to each child class and display their details.

Task 12: Hierarchical Inheritance – Vehicle → Car, Bike, Truck

Create a Vehicle base class and derive Car, Bike and Truck classes. Store common vehicle information and class-specific information, then display the details.

Task 13: Hierarchical Inheritance – Employee → Developer, Designer, Tester

Create an Employee base class and derive Developer, Designer and Tester classes. Store common employee details and role-specific information.

Task 14: Hybrid Inheritance – University Management System

Design a university management system using Person, Student, Teacher and GraduateStudent classes. Use an appropriate hybrid inheritance structure and display relevant details.

Task 15: Hybrid Inheritance – Company Employee Management

Create an employee management structure using Employee, Developer, Manager and TeamLead classes. Apply hybrid inheritance and display complete employee information.

Task 16: Bank Account System

Create an Account base class and derive SavingsAccount and CurrentAccount. Implement deposit, withdrawal and balance display operations using inheritance.

Task 17: Hospital Management

Create a Person base class and derive Doctor and Patient classes. Store and display appropriate information for doctors and patients using inheritance.

Task 18: School Management System

Create a Person base class and derive Student and Teacher classes. Manage student marks and teacher subject information using inheritance.

Task 19: Online Shopping System

Create Product, Electronics, Mobile and Laptop classes using inheritance. Store common and specific product details and display them.

Task 20: Vehicle Management System

Create Vehicle, Car, Bike and Truck classes using inheritance. Store common and specific properties and functions, then display complete vehicle details.

Suggested Order: Single → Multilevel → Multiple → Hierarchical → Hybrid → Real-world projects